Unit: Unit 0: Foundations of Algebra
Topic: Divisibility Division Algorithm

Name: ____________________
Date: ____________

Homework (Habanero)

Q1. A medical dosage requires 15 pills to be divided among 5 patients. How many pills will each patient receive, with a remainder of 0?
- (A) 3
- (B) 4
- (C) 5
- (D) 2
- (E) 1

Q2. A cell grows at a rate of 24 new cells per hour. If the cell growth is divided into 4 periods, how many new cells will each period have, with a remainder of 0?
- (A) 6
- (B) 8
- (C) 12
- (D) 4
- (E) 2

Q3. A hospital has 36 beds in a ward. If the beds are divided into 9 rooms, how many beds will each room have, with a remainder of 0?
- (A) 4
- (B) 5
- (C) 6
- (D) 3
- (E) 2

Q4. What is the remainder when 53 is divided by 5?
- (A) 3
- (B) 2
- (C) 1
- (D) 4
- (E) 0

Q5. What is the quotient when 48 is divided by 6?
- (A) 8
- (B) 7
- (C) 6
- (D) 5
- (E) 4

Q6. Is 27 divisible by 9?
- (A) Yes
- (B) No
- (C) Maybe
- (D) Possibly
- (E) Cannot determine

Q7. What is the remainder when 17 is divided by 3?
- (A) 2
- (B) 1
- (C) 0
- (D) 4
- (E) 5

Q8. Is 32 divisible by 2?
- (A) Yes
- (B) No
- (C) Maybe
- (D) Possibly
- (E) Cannot determine

Open Ended Questions (FRQ)

Q9. Divide 64 by 9 using long division. Express your answer with a remainder.

Q10. A doctor prescribes a patient 90 pills for 15 days. If the patient takes the same number of pills per day, how many pills will the patient take each day, with a remainder of 0? Show your work.

Chef's Tips

- Remind students to check their remainder for each division problem
- Encourage students to use real-life examples to understand the division algorithm
- Emphasize the importance of precise calculations when dealing with medical dosages